

ISO 19139 → JSON-LD Converter

Technical Documentation — LifeWatch ERIC spec v2.1

Version: 2.1 | Date: July 14, 2026

Scope: Workflows, VRES, Services (Action / HowTo / CreativeWork)

What changed in v2.1: provider hardcoded to LifeWatch ERIC block on all record types; empty affiliation now included per sheet spec.

1. Overview

This Google Colab notebook converts **ISO 19139 XML** metadata records into **JSON-LD** documents conforming to the LifeWatch ERIC mapping spec. Unlike the EML converter (which always produces Dataset), this notebook **auto-detects the record type** and maps to the appropriate schema.org type.

1.1 Record type auto-detection

ISO 19139 signal	JSON-LD @type	Detection logic
hierarchyLevel=service + registeredAction	Action	srv:SV_OperationMetadata or gmd:LW_Service present
hierarchyLevel=service + Marco-Bolo keyword + no I/O	HowTo	"marco-bolo" in title or keywords
hierarchyLevel=service + no Marco-Bolo keyword + no I/O	CreativeWork	Default for services
hierarchyLevel=application or "workflow/VRE" in title/keywords	Workflow or VRE	Workflow or VRE signals
Default (any other)	CreativeWork	Fallback

1.2 Dependencies

Package	Purpose
lxml	XML parsing of ISO 19139 documents
tabulate	Formatted mapping-report tables

1.3 Entry Point

In Google Colab, run() with no arguments triggers a file-upload dialog. Record type is auto-detected. To override:

```
result = run(xml_path='/content/file.xml', record_type='Action') # Options:
'Workflow', 'VRE', 'Action', 'HowTo', 'CreativeWork'
```

2. What's New in v2.1

2.1 provider — hardcoded LifeWatch ERIC block

Every record type now includes the hardcoded LifeWatch ERIC provider block, consistent with the EML converter and the LifeWatch ERIC spec:

```
"provider": { "@type": "Organization", "name": "e-Science and Technology European
Infrastructure for Biodiversity and Ecosystem Research", "alternateName": "LifeWatch
ERIC", "identifier": "https://ror.org/04c04g438", "url": "https://www.lifewatch.eu",
"email": "communications@lifewatch.eu" }
```

2.2 Empty affiliation fix

When a contact's <organisationName> element is present but empty, the sheet spec requires "affiliation": {"@type": "Organization", "name": ""}. Previously the affiliation was silently omitted. Fixed in v2.1.

Before (v1 — missing affiliation for empty org):

```
{"@type": "Person", "name": "Ben Hamner", "email": ""}
```

After (v2.1 — affiliation always present):

```
{"@type": "Person", "name": "Ben Hamner", "affiliation": {"@type": "Organization",  
"name": ""}, "email": ""}
```

3. Field Mapping by Record Type

3.1 Workflow and VRE → CreativeWork

ISO 19139 Field	JSON-LD Property	Notes
fileIdentifier	@id	https://metadatalogue.lifewatch.eu/srv/api/records/{UUID}
fileIdentifier	url	https://metadatalogue.lifewatch.eu/srv/eng/catalog.search#/m
title	name	—
abstract	description	—
CI_Date (creation)	dateCreated	—
CI_Date (publication)	datePublished	Empty string if element present but empty
CI_Date (revision)	dateModified	Empty string if element present but empty
MD_ProgressCode	creativeWorkStatus	DefinedTerm with PSO URI (see status map)
pointOfContact	creator	Person/Organization — all contacts included
useLimitation	license	Pass through as-is; use URL when available
keyword	keywords	Plain string array
distributionInfo/URL	sameAs	All URLs — string if 1, array if multiple
(hardcoded)	provider	LifeWatch ERIC organisation block — always

3.2 Service → Action

Used when the service has registered input/output datasets in the catalogue (srv:SV_OperationMetadata or gmd:LW_Service).

ISO 19139 Field	JSON-LD Property	Notes
fileIdentifier	@id + url	Same pattern as above
title	name	—
abstract	description	—
CI_Date (publication or creation)	startTime + endTime	Same value for both; publication date takes priority
MD_ProgressCode	creativeWorkStatus	DefinedTerm with PSO URI
pointOfContact	agent	agent (not creator) for Action type
useLimitation	(not mapped)	License not mapped for Action per spec
keyword	(not mapped)	Keywords not mapped for Action per spec
distributionInfo/URL	sameAs	Last URL only (the runnable workflow/NaaVRE link)
(hardcoded)	provider	LifeWatch ERIC organisation block — always

3.3 Service → HowTo

Used when the service is part of Marco-Bolo and has no registered I/O.

ISO 19139 Field	JSON-LD Property	Notes
fileIdentifier	@id + url	—
title	name	—
abstract	description	—
CI_Date (creation/publication/revision)	dateCreated / datePublished / dateModified	—
MD_ProgressCode	creativeWorkStatus	DefinedTerm with PSO URI
pointOfContact	creator	Person/Organization — all contacts
useLimitation	license	Use URL when available
keyword	keywords	Plain string array
distributionInfo/URL	sameAs	All URLs
(hardcoded)	provider	LifeWatch ERIC organisation block — always

3.4 Service → CreativeWork

Used when the service has no registered I/O and is not part of Marco-Bolo. Same field mapping as HowTo — only @type differs.

3.5 MD_ProgressCode → creativeWorkStatus mapping

ISO code	name	PSO URI
completed	Published	http://purl.org/spar/psa/published
onGoing	Published	http://purl.org/spar/psa/published
underDevelopment	Draft	http://purl.org/spar/psa/draft
historicalArchive	Archived	http://purl.org/spar/psa/archived
obsolete	Archived	http://purl.org/spar/psa/archived

4. Complete Mapping Summary

ISO 19139 Field	Status	Workflow / VRE	Action	HowTo / CreativeWork
fileIdentifier	OK	@id + url	@id + url	@id + url
title	OK	name	name	name
abstract	OK	description	description	description
CI_Date creation	OK	dateCreated	—	dateCreated
CI_Date publication	OK	datePublished	startTime + endTime	datePublished
CI_Date revision	OK	dateModified	—	dateModified
MD_ProgressCode	OK	creativeWorkStatus	creativeWorkStatus	creativeWorkStatus
pointOfContact	OK	creator	agent	creator
useLimitation	OK / NOT MAPPED	license	not mapped	license
keyword	OK / NOT MAPPED	keywords[]	not mapped	keywords[]
distributionInfo/URL	OK	sameAs (all URLs)	sameAs (last URL)	sameAs (all URLs)
provider (LifeWatch ERIC)	OK	hardcoded block	hardcoded block	hardcoded block

5. Architecture

Component	Role
Constants / NS	JSONLD_CONTEXT, CATALOGUE_HOST, STATUS_MAP, UUID_RE, NS namespaces
_xp / _text / _texts / _attr	XPath helper utilities
parse_contact()	Build Person/Organization node from CI_ResponsibleParty
get_dates()	Extract creation/publication/revision dates from CI_Date elements
get_distribution_urls()	Collect all online URLs from distributionInfo
detect_record_type()	Auto-detect Action/HowTo/CreativeWork from hierarchyLevel + keywords
ISO19139toJsonLD class	_map_* methods + convert() + save() + print_loss_report()
run()	Top-level orchestrator (upload → convert → save → preview → download)

5.1 Conversion pipeline

_map_id() — Sets @id and url from fileIdentifier UUID

_map_basic() — Maps title → name, abstract → description

_map_dates() — Maps dates differently per type: dateCreated/Published/Modified or startTime+endTime

_map_status() — Maps MD_ProgressCode → creativeWorkStatus DefinedTerm with PSO URI

`_map_contacts()` — Maps pointOfContact → creator or agent; injects LifeWatch ERIC provider
`_map_license()` — Maps useLimitation → license (skipped for Action)
`_map_keywords()` — Maps keywords as plain strings (skipped for Action)
`_map_distribution()` — Maps distributionInfo URLs → sameAs (last URL only for Action)

6. Usage Guide

6.1 Google Colab

Runtime → Restart runtime → Runtime → Run all, then upload your ISO 19139 XML. Record type is auto-detected from the XML content.

6.2 Programmatic usage

```
# Auto-detect (recommended) result = run(xml_path='/path/to/record.xml') # Override
record type result = run(xml_path='/path/to/record.xml', record_type='Action') #
Direct class usage conv = ISO19139toJsonLD('/path/to/record.xml') doc =
conv.convert(record_type='HowTo') conv.save('record.jsonld') conv.print_loss_report()
```

6.3 Mapping report status codes

Status	Meaning
OK	Field successfully mapped
PARTIAL	Field attempted but only partially handled
LOST	Field absent in the source XML

7. Version History

Version	Change
v2.1	provider hardcoded to LifeWatch ERIC block on all record types; empty affiliation fix.
v2.0	Auto-detection of record type added (Workflow/VRE/Action/HowTo/CreativeWork).
v1.0	Initial implementation covering Workflow, VRE, Action, HowTo, CreativeWork.

Generated from ISO19139_to_JsonLD_v2.1.ipynb — LifeWatch ERIC spec v2.1. External validation:
<https://validator.schema.org/>